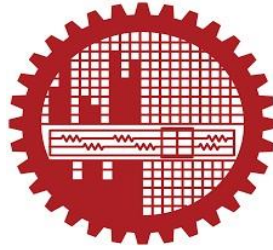


Bangladesh University of Engineering and Technology



Department of Materials and Metallurgical Engineering

Course No: MME 308

Course Title: Materials Laboratory I

Experiment No: 03

Experiment Title: Design of polymer matrix composite for car clutch disk

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Objective:

1. To prepare fiber-reinforced polymer composite by hand layup and hot compression molding technique.
2. To understand the behavior of reinforcements on the polymer composites by hardness, tensile, flexural tests, impact test, and FTIR.

Equipment:

- Asbestos
- Pipes (6" and 10" diameter)
- Epoxy resin
- Sample car clutch (Hiace car) [A replica will be carefully created with it as the primary model.]
- Nose mask
- Syringe
- Plastic sheet
- Required oily substance

Experimental procedure:

Collection of raw material and primary model:

1. First, we collected a primary model of a car clutch (Hiace). We collected it to make a replica.



Figure 1: Car clutch of Hiace

2. Then, we measured the inner diameter, outer diameter, and thickness of the polymer part of the car clutch (inner diameter = 6", outer diameter = 10", and thickness = 10").
3. We bought two pipes with diameters of 6 inches and 10 inches.
4. We bought asbestos as our polymer.

Preparation of polymer design:

1. We cut the plastic sheet and shaped it using the pipe.
2. We adjusted the sheet with the pipe.
3. We prepared a solution with resin and an oily substance.
4. We filled the space between the two pipes (the 4" gap) placed on the plastic sheet with asbestos and the solution.



*Figure 2: Placement of Resin and Solution
Between the Inner and Outer Pipes*

5. We waited for 24 hours for the solution to dry.
6. After that, we obtained the final product and performed the characterization tests (tensile, hardness, impact, flexure, FTIR) as follows.

Data Collection:

Tensile test:

Gauge length: 12.5 mm (Before fracture)

Gauge length: 52.5 mm (After fracture)

Hardness:

Using Shore-D hardness tests we got 5 values at different parts of the car clutch:

43, 62, 54, 51.5, and 54.5 HD

Impact:

The impact test result of our sample is 0.2 J

Flexural test:

Specimen, length = 81.35 mm; width = 16.1 mm; thickness = 4 mm

And gap between the two supported pieces during the test was 64 mm

Calculations:

Tensile Test:



Figure 3: Tensile test specimen

Dimensions of our tensile specimen:

Length, $l = 12.5$ mm (Before fracture) & 52.5 mm (After fracture)

Width, $b = 25$ mm

Thickness, $t = 6$ mm

Cross-sectional area, $A = 25 \times 6 \times 10^{-6} \text{ m}^2 = 1.5 \times 10^{-4} \text{ m}^2$

From the graph, Load = 0.076 kN

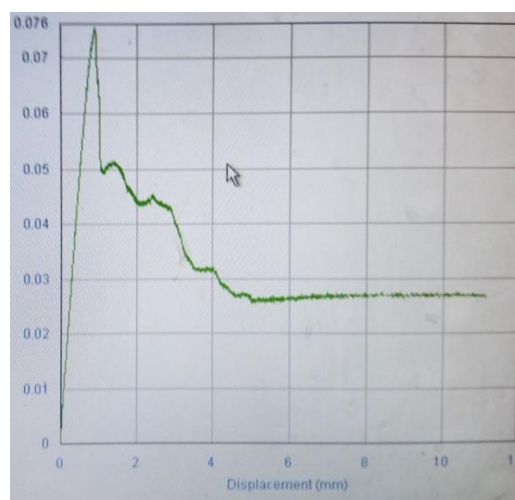


Figure 4: Tensile Test Graph

$$\text{Tensile Strength} = \frac{\text{Load}}{\text{Cross-sectional area}} = \frac{0.076 \times 10^3 \text{ N}}{1.5 \times 10^{-4} \text{ m}^2} = 0.51 \text{ MPa}$$

$$\% \text{Elongation} = \frac{52.5 - 12.5}{12.5} \times 100\% = 320\%$$

Hardness test:

$$\text{Average hardness} = \frac{43 + 62 + 54 + 51.5 + 54.5}{5} \text{ HB} = 53 \text{ HB}$$

Flexural test:

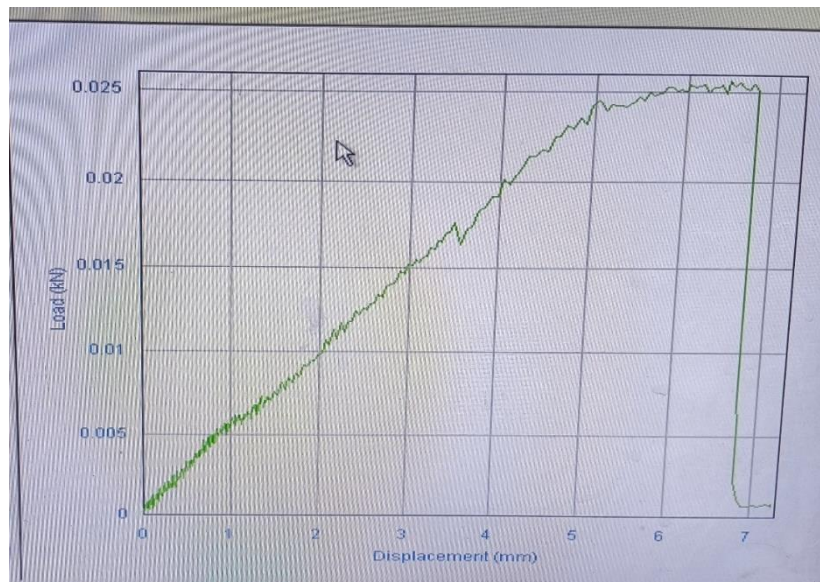


Figure 5: Flexural test graph

By analyzing this graph,

Applied load at the fracture point,
 $P = 0.02566 \text{ kN}$ (at displacement = 6.73545 mm) the gap between two supported pieces during the test, $L = 64 \text{ mm}$

Width of the specimen, $b = 16.1 \text{ mm}$

Thickness of the specimen, $d = 4 \text{ mm}$

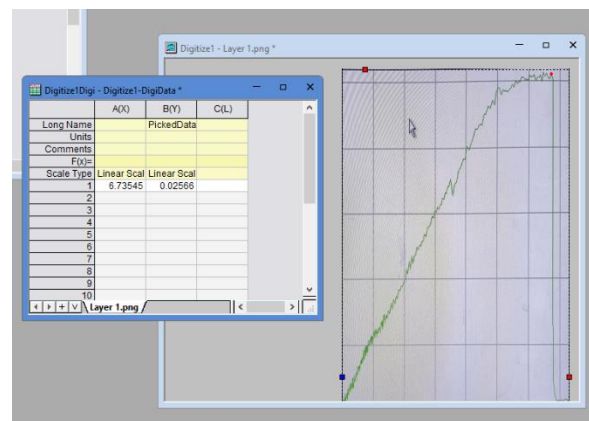


Figure 6: Use of Origin plus to find the peak

We know,

$$\text{Flexural strength, } \sigma_y = \frac{3PL}{2bd^2} = 9.56 \text{ MPa}$$

Results:



Figure 7: Our Final product (The polymer design of the car clutch)

Our final product is a replica of a Hiace car clutch. We made it using asbestos, resin, and other substances with the help of our lab instructor. Afterward, we took different parts of our final sample and conducted various tests to evaluate its capability, strength, composite-matrix ratio, and applicability to the real clutch. The test results are presented below:

Tensile test:

From calculations, Tensile Strength = 0.51 MPa

%Elongation = 320%

Hardness test:

From calculations, Hardness of our sample= 53HB

Impact test:

The impact test result of our product is 0.2J

Flexural test:

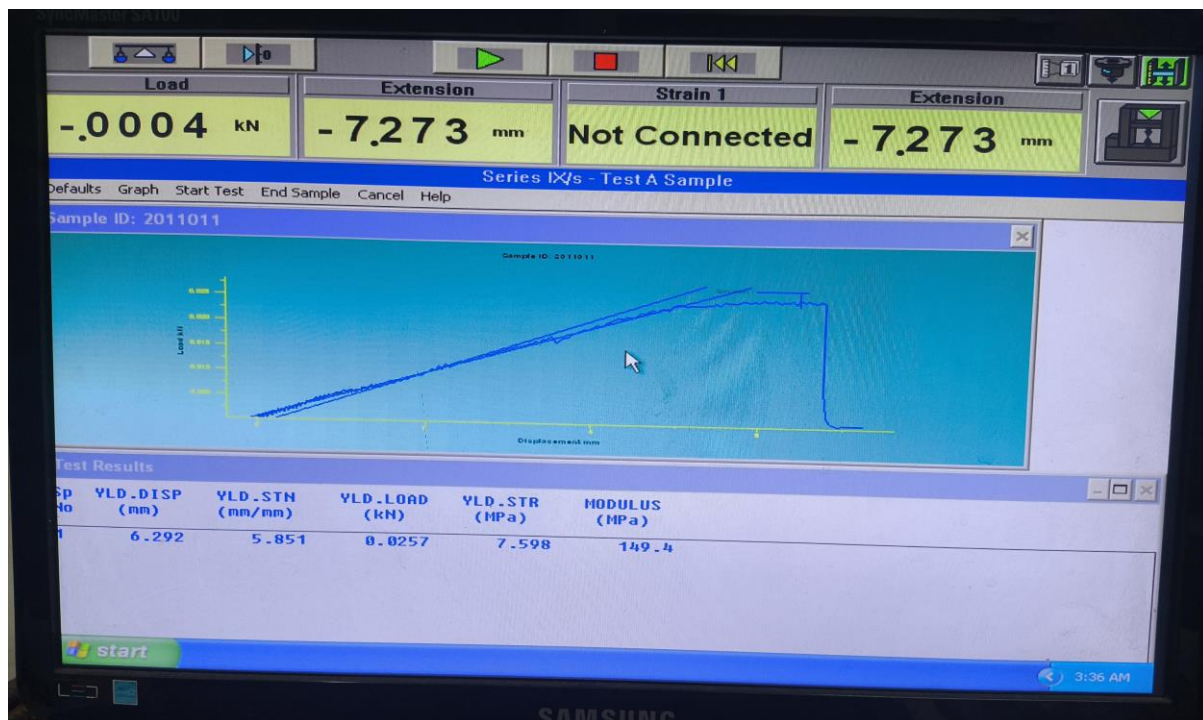


Figure 8: Flexural Test result

From this graph-

Yield Displacement = 6.292 mm

Yield strain = 5.851 mm/mm

Yield Load = 0.0257 kN

Yield Stress = 7.598 MPa

Young's Modulus/Flexural Modulus = 149.4 MPa

From calculations, we get that Flexural strength = 9.56 MPa

FTIR Test:

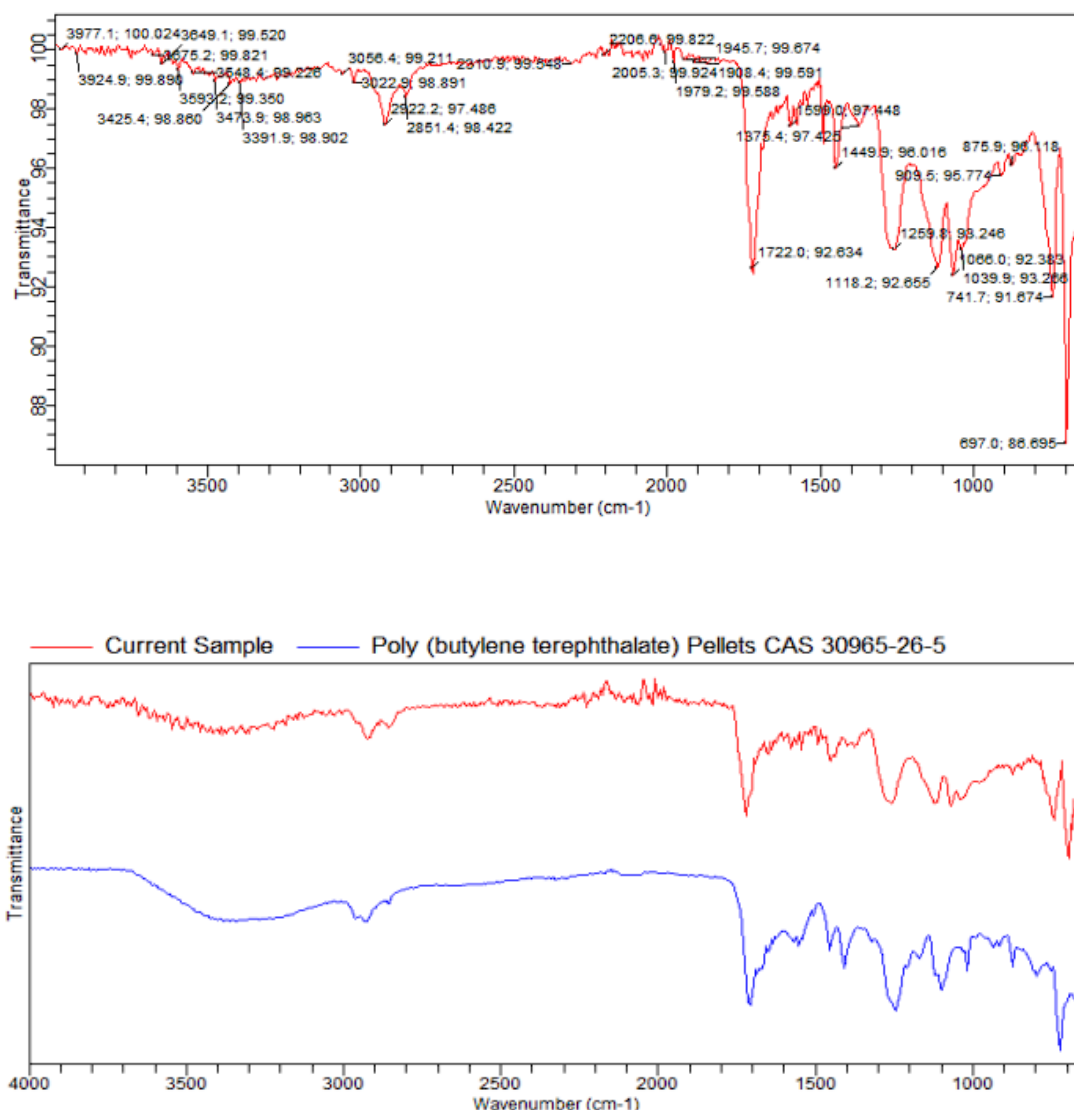


Figure 9: FTIR test of our prepared product

Analysis overview: The samples were analyzed using an Agilent system with a Happ-Genzel apodization method, covering a spectral range of 4000 to 650 cm⁻¹. The system performed well, and data quality was considered good. To ensure a balanced measurement, background, and sample scans were set to 32 each.

Material Identification:

1. **Poly (butylene terephthalate) Pellets (CAS 30965-26-5):** The sample showed a high match (0.83605) to Poly (butylene terephthalate) in the polymer library.
2. **Vinyl chloride_vinyl acetate_hydroxylpropyl acrylate terpolymer (CAS 41618-91-1):** Our sample had a slightly lower correlation (0.78387) with this material. We could consider it as the second place compared to the previous one and can say, it has a lower amount of Vinyl chloride_vinyl acetate_hydroxylpropyl acrylate terpolymer.

The rest of the materials identified had varying degrees of matches, including elastomers and other copolymers, indicating a diverse composition in the analyzed sample set.

Discussion:

The lab experiment aimed to create a fiber-reinforced polymer composite using a replica of a Hiace car clutch as a model. Various tests, such as tensile, hardness, impact, and flexural, were performed on the composite to assess its mechanical properties. While this was a controlled lab setting, actual car clutch manufacturing follows an industrial process that includes high-precision engineering, rigorous testing, and the use of specific materials suited for high-stress automotive components.

Materials Comparison:

1. **Lab Work:** In the experiment, we used **asbestos** as the primary reinforcing material in combination with **epoxy resin**. Asbestos was chosen due to its thermal resistance and strength, which are useful characteristics in composites meant for high-heat applications.
2. **Actual Car Clutch:** In real-world car clutches, modern composites often use **aramid fibers (such as Kevlar)** or **carbon fibers** for reinforcement instead of asbestos. Asbestos is not used due to health concerns. Kevlar offers high tensile strength and is lightweight, making it an ideal choice for automotive components subjected to friction and heat.
3. **Comment:** The material choice in the lab was more experimental and less suited for real-world applications due to the health risks of asbestos and the superior performance of modern materials in the actual clutch manufacturing process.[1]

Manufacturing Process Comparison:

1. **Lab Work:** The lab process involved hand layup and hot compression molding. The composite material was molded into the desired clutch shape by filling a space between two pipes with the resin-asbestos solution. This method allowed for manual control over the shape and material distribution.
2. **Actual Car Clutch:** In industry, clutches are manufactured using high-precision **injection molding** or **compression molding** techniques under strict quality control measures. **CNC machining** and **automated processes** ensure that the clutch components meet exact specifications, and automation reduces human error, ensuring consistency in large-scale production.
3. **Comment:** While the lab process provided insights into material behavior and manual fabrication techniques, actual clutch manufacturing is more refined, with automated processes enhancing precision and product quality.[2]

Tensile Strength:

1. **Lab Work:** The tensile strength of the composite in the lab was **0.51 MPa** and the %elongation was 320%. This relatively low tensile strength indicates that the material created in the lab is not ideal for handling the high stresses experienced by a car clutch in operation.
2. **Actual Car Clutch:** A typical automotive clutch requires materials with tensile strengths in the range of **500-600 MPa**, far exceeding the tensile properties observed in the lab. Clutch discs are often made of heat-treated steel and friction material capable of withstanding high forces during acceleration and braking.
3. **Comment:** The tensile strength in the lab product was significantly lower than that of actual clutches, limiting the application of the lab material for real-world automotive use.[3]

Hardness:

1. **Lab Work:** The hardness of the composite was measured using a **Shore-D** test, yielding an average value of **53 HD**. This indicates moderate resistance to surface deformation, suitable for understanding basic polymer behavior but not sufficient for high-stress applications.
2. **Actual Car Clutch:** Car clutches require materials with high surface hardness to resist wear under continuous friction. A clutch friction material has hardness ratings that range from **80-90 HD** on the Shore-D scale. The higher hardness ensures durability and prolonged service life under harsh operating conditions.
3. **Comment:** The lab material's hardness was lower, which means it would likely wear out faster in real-world applications.[4]

Impact Resistance:

1. **Lab Work:** The impact test result for the lab composite was **0.2 J**, which is relatively low. This means the material would not perform well in dynamic conditions where it might face sudden loads or shocks.
2. **Actual Car Clutch:** In real-world applications, clutch discs must have significantly higher impact resistance, as they undergo repeated engagement and disengagement under heavy loads. Impact resistance in the automotive industry is generally optimized through advanced materials that absorb and dissipate energy without cracking.
3. **Comment:** The low impact resistance of the lab material indicates its unsuitability for high-performance, dynamic applications like those required in car clutches.

Flexural Strength:

1. **Lab Work:** The flexural strength of the composite was calculated as **9.56 MPa**, with **Young's modulus** of **149.4 MPa**. These values suggest that while the material can withstand some bending stress, it lacks the rigidity needed for high-stress automotive parts.
2. **Actual Car Clutch:** The materials used in real clutches have flexural strengths ranging from **80 to 100 MPa** or more, allowing them to resist bending and flexing during

operation. The high modulus ensures that the clutch maintains its shape under mechanical stresses.

3. **Comment:** The lab material's low flexural strength means it would not maintain its structural integrity under the loads experienced in a car clutch.[5]

FTIR Results:

1. **Lab Work:** The FTIR analysis identified **Poly(butylene terephthalate)** as the primary polymer in the composite. This material is generally known for its thermal resistance and chemical stability but lacks the mechanical strength required for automotive components.
2. **Actual Car Clutch:** Actual clutches use a combination of steel and high-performance friction materials (such as Kevlar or carbon composites), which are optimized for both mechanical and thermal performance, offering the strength and durability required in high-stress conditions.
3. **Comment:** Therefore, the chemical composition indicates that we can use it for general purposes rather than for the special needs of a car clutch.[6]

A comparison table between our product and an actual car clutch:-

Aspect	Lab Product	Actual Car Clutch	Comments
Reinforcing Material	Asbestos with epoxy resin	Aramid fibers (Kevlar) or carbon fibers	Asbestos is thermally resistant but is phased out due to health risks; Kevlar and carbon fibers offer superior strength and are safer.
Manufacturing Process	Hand layup, hot compression molding	High-precision injection/compression molding	Lab methods are manual, while actual clutches use automated, high-precision processes for consistency and durability.
Tensile Strength	0.51 MPa, 320% elongation	500-600 MPa	Lab product has significantly lower tensile strength, unsuitable for the high-stress environment of a car clutch.
Hardness	53 HD (Shore-D)	80-90 HD (Shore-D)	Lab composite has a lower hardness, making it prone to faster wear compared to actual clutches.
Impact Resistance	0.2 J	High, optimized for repeated engagement	Low impact resistance in the lab product indicates poor performance in dynamic automotive environments.

Flexural Strength	9.56 MPa, Young's modulus: 149.4 MPa	80-100 MPa	The lab product lacks the rigidity and strength needed to maintain its shape under high mechanical stresses.
FTIR Composition	Poly(butylene terephthalate)	Steel and high-performance friction materials	The lab composite lacks the mechanical strength and durability provided by steel and advanced friction materials in real clutches.

Conclusion:

The lab experiment provided valuable insights into the behavior of fiber-reinforced polymer composites and allowed for an understanding of basic material properties. However, when compared to actual car clutch manufacturing, the lab materials and processes fall short in terms of mechanical strength, hardness, and impact resistance. The use of asbestos in the lab, while functional, is outdated and less effective than modern materials like Kevlar or carbon fibers. Actual car clutches are produced through more advanced manufacturing techniques and with better materials, ensuring they can withstand the demanding conditions of automotive use.

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